Admissible level of deterministic noise that is orthogonal to the randomization on average over a Markov state

Working problem statement
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1. Context

We consider gradient-free minimization of a convex function f on a ball of radius R in $\mathbb{R}^d$ by the two-point method

$$\hat{g}(x) = \frac{d}{2\gamma}\big(O(x+\gamma e) - O(x-\gamma e)\big)\, e, \qquad e \sim \mathrm{Unif}(S^{d-1}),$$

where O is the function value with an error and γ is the smoothing radius.

Adversarial model. The oracle is $O(x) = f(x) + \delta(x)$ with $|\delta(x)| \le \Delta$; the error is fixed (a repeated query returns the same answer) and may depend on x arbitrarily. The noise then adds a *bias* of order $\sqrt{d}\,\Delta/\gamma$ to the gradient estimate, and accuracy ε requires $\Delta \lesssim \varepsilon\gamma/(\sqrt{d}R)$. The maximal admissible noise level in this model is now known up to logarithmic factors [2, 3]:

$$\Delta^{\mathrm{adv}}_{\max} \asymp \frac{\varepsilon^2}{\sqrt{d}\,MR} \qquad (M\text{-Lipschitz } f),$$

$$\Delta^{\mathrm{adv}}_{\max} \asymp \frac{\varepsilon^{3/2}}{\sqrt{d}\,\sqrt{L}\,R} \qquad (L\text{-smooth } f, \text{ Lipschitz with } M \sim \sqrt{L\varepsilon}).$$

Orthogonality to the randomization (survey [1], §4.4):

$$\mathbb{E}_e\big[\delta(x+\gamma e)\, e\big] = 0 \qquad \forall x. \tag{1}$$

Under (1) the noise causes no bias and enters only the variance of the estimate; heuristically the remaining condition is $d\Delta^2/\gamma^2 \lesssim M^2$, i.e. $\Delta \lesssim M\gamma/\sqrt{d}$ (for Lipschitz f with $\gamma \sim \varepsilon/M$ this is $\varepsilon/\sqrt{d}$, a factor MR/ε above the adversarial threshold). In practice, however, condition (1) almost never holds pointwise, at every x. The typical situation is that it holds *on average over some state* that changes over time.

Markovian noise. It is shown in [4] that for gradient-free methods with Markovian stochastic noise (mixing time τ, a chain that does not depend on the iterates) the cost of correlation is additive: the oracle complexity grows as $d + \tau$ rather than $d\tau$. A deterministic error is handled there only as a bounded adversarial one.

2. Noise class

Definition 1 (Noise orthogonal to the randomization on average over a Markov state)**.** At step k the oracle returns $O(x, s_k) = f(x) + \delta(x, s_k)$, and both queries of the two-point estimate are made in the same state s_k. We assume:

(i) $|\delta(x, s)| \le \Delta$ for all x, s;

(ii) $\{s_k\}$ is a uniformly geometrically ergodic Markov chain with stationary law π and mixing time τ;

(iii) condition (1) holds under the stationary law:

$$\mathbb{E}_{s\sim\pi}\,\mathbb{E}_e\big[\delta(x+\gamma e, s)\, e\big] = 0 \qquad \forall x; \tag{2}$$

for a fixed s the noise may be biased.

The class is denoted $\mathcal{N}(\Delta, \tau, \gamma)$.

Laziness. The main mechanism of slow mixing is the *laziness* of the chain: with probability α the state does not change at a step,

$$P_{\text{lazy}} = \alpha I + (1-\alpha)P, \qquad \text{mean holding time } \tfrac{1}{1-\alpha}, \qquad \tau \gtrsim \tfrac{1}{1-\alpha}.$$

While a state s is held, the noise $\delta(\cdot, s)$ is the same, and its bias for fixed s repeats instead of averaging out: over $\sim \tau$ steps the deterministic error behaves like an adversarial one. Condition (2) starts to "work" only on the scale of the mixing time. In the limiting case $\alpha = 1$ on part of the states (absorption) the chain does not mix at all.

Special cases.

- $\tau = 1$ (independent states): the stochastic axiomatics of condition (1);
- an external Markov data chain: the model of [4] with an additional deterministic component;
- *rounding* of function values to the grid $u\mathbb{Z}$: the state is the position of $f(x)$ on the grid (the offset). It is moved both by the data and by the method itself: on a step where the two rounded values coincide (a "tie") the point does not move, and with the same data the offset repeats. This is a lazy chain with holding probability $\alpha = P_{\text{tie}}(\omega, r) = 2\Phi\big((\tfrac{1}{2} - |\omega|)/r\big) - 1$, where ω is the offset and $r = \gamma\|\nabla f\|/u$; the kernel of the chain depends on the iterates, and for $r \ll 1$ and unchanged data the states in the middle of a cell are practically absorbing. For this case condition (2) has been verified exactly: for f affine on the stencil and Gaussian directions, $\mathbb{E}_e \hat{g} = \kappa(c, r)\nabla f$, where κ is a wrapped Gaussian of the offset c with width $r = \gamma\|\nabla f\|/u$, and $\int_0^1 \kappa\, dc = 1$.

3. Problem

Goal

For the class $\mathcal{N}(\Delta, \tau, \gamma)$ and the classes of Lipschitz and smooth convex functions, determine

$$\Delta_{\max}(\varepsilon, d, \tau, \gamma) = \sup\big\{\Delta : \ \exists\, \mathcal{A} \ \ \forall f \ \ \forall \delta \in \mathcal{N}(\Delta, \tau, \gamma) : \ \ \mathbb{E} f(\hat{x}_{\mathcal{A}}) - f^* \le \varepsilon\big\},$$

where $\mathcal{A}$ is an algorithm with a polynomial number of oracle calls, and obtain two-sided bounds on it:

- **upper bound**: a construction of a hard function and an admissible noise for which no algorithm reaches accuracy ε (the existence side, following the scheme of [2]);
- **lower bound**: a two-point method with explicit parameters (radius, batch, step) that reaches ε for the given Δ.

Compare the resulting thresholds with the adversarial $\Delta_{\max}^{\text{adv}}$.

Expected picture.

$$\Delta_{\max}^{\text{adv}} \ \lesssim \ \Delta_{\max}(\varepsilon, d, \tau, \gamma) \ \lesssim \ \frac{M\gamma}{\sqrt{d}}.$$

While the chain mixes, the threshold is determined by the variance and deteriorates as τ grows, i.e. as the laziness α grows: the bias accumulates over the holding time. When mixing breaks down ($\alpha \to 1$, absorbing states), the adversarial regime returns. A separate question is how the threshold depends on laziness generated by the method itself (rounding) rather than by an external chain.

Plan of work.

1. $\tau = 1$: rewrite the bound of §4.4 [1] in terms of the variance; construct a hard function for the class with condition (1) in stochastic form.
2. External chain, $\tau > 1$: transfer the technique of [4] (a direction per sample, multilevel Monte Carlo) to the component that is biased for a fixed state; obtain the dependence of the threshold on τ.
3. Lazy chain that depends on the iterates (rounding): relation between laziness and the tie probability, mixing conditions, absorbing states, comparison with the adversarial threshold.

Notation. The smoothing radius is γ here and μ in the experiments of this repository; they denote the same quantity.

References

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[2] D. Pasechnyuk-Vilensky, I. Pavlov, M. Takáč, A. Gasnikov. Information-theoretic upper bounds for deterministic noise in zeroth-order convex optimization. arXiv:2306.16371v5, 2026.

[3] A. Risteski, Y. Li. Algorithms and matching lower bounds for approximately-convex optimization. NeurIPS, 2016.

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